Data375Project

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Data sets

```r
data1 <- read.csv("/Users/avamizrahi/Downloads/Student_Loan_Debt.csv")

data2 <- read.csv("/Users/avamizrahi/Downloads/College_Rankings.csv")

head(data1)
```

```
##   index OPE.ID                             School       City State Zip.Code
## 1     0 001051              UNIVERSITY OF ALABAMA TUSCALOOSA    AL    35487
## 2     1 001052 UNIVERSITY OF ALABAMA AT BIRMINGHAM BIRMINGHAM    AL    35294
## 3     2 001009                  AUBURN UNIVERSITY     AUBURN    AL    36849
## 4     3 001057        UNIVERSITY OF SOUTH ALABAMA     MOBILE    AL    36688
## 5     4 001047                    TROY UNIVERSITY       TROY    AL    36082
## 6     5 001036                 SAMFORD UNIVERSITY BIRMINGHAM    AL    35229
##         School.Type  Loan.Type Recipients X..of.Loans.Originated
## 1            Public Subsidized       7594                   7622
## 2            Public Subsidized       4272                   4352
## 3            Public Subsidized       4504                   4504
## 4            Public Subsidized       3438                   3487
## 5            Public Subsidized       4334                   4355
## 6 Private-Nonprofit Subsidized        638                    638
##   X..of.Loans.Originated.1 X..of.Disbursements X..of.Disbursements.1
## 1                 33040862                7622              16774054
## 2                 18590061                4352               9058344
## 3                 19474542                4535              10142490
## 4                 15061071                3487               7557853
## 5                 18467508                4355               7395839
## 6                  2725011                 638               1412105
```

```r
head(data2)
```

```
##                          College.Name Adjusted.Rank Tuition
## 1                  Princeton University             1   56010
## 2                   Columbia University             2   63530
## 3                    Harvard University             2   55587
## 4 Massachusetts Institute of Technology             2   55878
## 5                       Yale University             5   59950
## 6                   Stanford University             6   56169
```

```
##   Enrollment.Numbers
## 1               4773
## 2               6170
## 3               5222
## 4               4361
## 5               4703
## 6               6366
```

Combining the 2 datasets

```r
data1$School <- tolower(data1$School)
data2$School <- tolower(data2$College.Name)

College_Data <- merge(data1, data2, by = "School")
head(College_Data)
```

```
##              School index OPE.ID       City State Zip.Code       School.Type
## 1 american university   258 001434 WASHINGTON    DC    20016 Private-Nonprofit
## 2 american university 13080 001434 WASHINGTON    DC    20016 Private-Nonprofit
## 3 american university 34450 001434 WASHINGTON    DC    20016 Private-Nonprofit
## 4 american university 25902 001434 WASHINGTON    DC    20016 Private-Nonprofit
## 5 american university  8806 001434 WASHINGTON    DC    20016 Private-Nonprofit
## 6 american university 21628 001434 WASHINGTON    DC    20016 Private-Nonprofit
##                Loan.Type Recipients X..of.Loans.Originated
## 1             Subsidized       1876                   1878
## 2            Parent Plus        365                    369
## 3                  Total      43959                  44190
## 4                  Total      12518                  12584
## 5 Unsubsidized - Graduate       2259                   2260
## 6                  Total       7197                   7231
##   X..of.Loans.Originated.1 X..of.Disbursements X..of.Disbursements.1
## 1                  8499687                1878               4447220
## 2                 11692877                 369               6425163
## 3                512364727               44190             310701818
## 4                140137683               12584              84347404
## 5                 37193281                2260              21028272
## 6                 74318685                7231              44397312
##          College.Name Adjusted.Rank Tuition Enrollment.Numbers
## 1 American University            79   51334               7953
## 2 American University            79   51334               7953
## 3 American University            79   51334               7953
## 4 American University            79   51334               7953
## 5 American University            79   51334               7953
## 6 American University            79   51334               7953
```

```r
College_Data <- na.omit(College_Data)
Ranking_Tuition <- College_Data

library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
colnames(College_Data)
```

```
##  [1] "School"                   "index"
##  [3] "OPE.ID"                   "City"
##  [5] "State"                    "Zip.Code"
##  [7] "School.Type"              "Loan.Type"
##  [9] "Recipients"               "X..of.Loans.Originated"
## [11] "X..of.Loans.Originated.1" "X..of.Disbursements"
## [13] "X..of.Disbursements.1"    "College.Name"
## [15] "Adjusted.Rank"            "Tuition"
## [17] "Enrollment.Numbers"
```

```r
colnames(Ranking_Tuition) <- c(
  "School", "index", "OPE.ID",
  "City", "State", "Zip.Code",
  "School.Type", "LoanType", "Recipients",
  "XLoansOriginated", "PLoansOriginated", "XDisbursements",
  "PDisbursements", "CollegeName", "AdjustedRank",
  "Tuition", "EnrollmentNumbers"
)
```

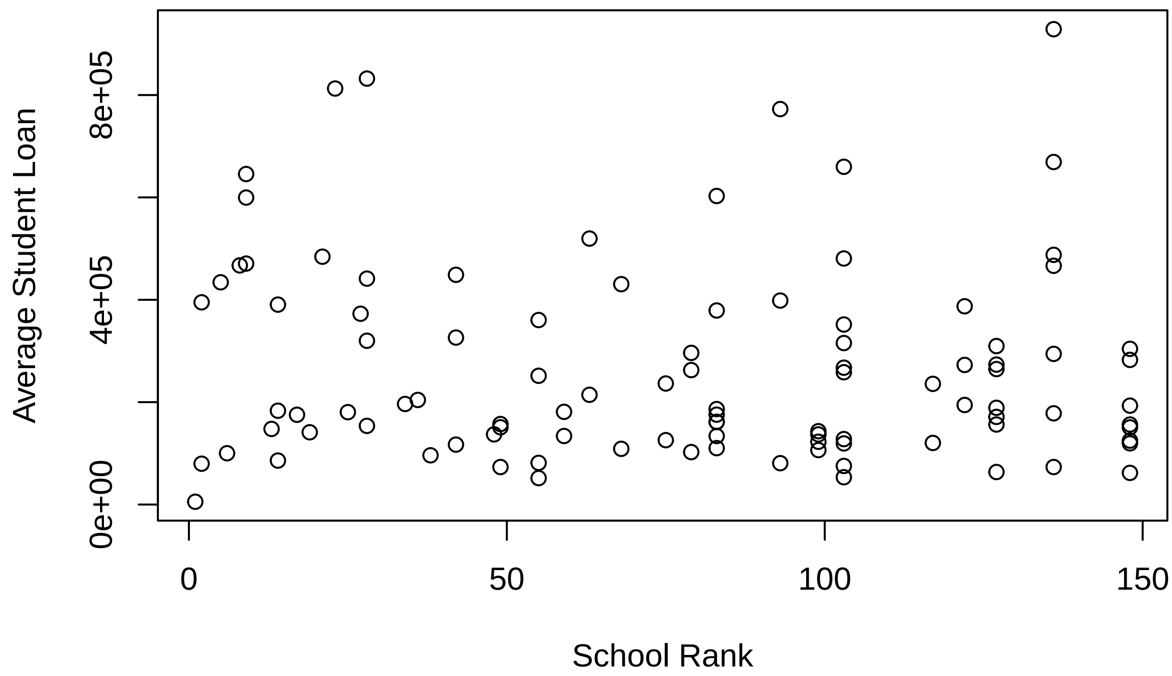
```r
Ranking_Tuition <- Ranking_Tuition %>%
  group_by(School) %>%
  summarize(
    Tuition = mean(Tuition),
    Rank = mean(AdjustedRank),
    Enrollment = mean(EnrollmentNumbers),
    Num_Loans = sum(XLoansOriginated),
    Total_Loan_amt = sum(PLoansOriginated),
    Num_Disbursements = sum(XDisbursements),
    Total_Disbursements_amt = sum(PDisbursements)
  )
colnames(Ranking_Tuition)
```

```
## [1] "School"                  "Tuition"
## [3] "Rank"                    "Enrollment"
## [5] "Num_Loans"               "Total_Loan_amt"
## [7] "Num_Disbursements"       "Total_Disbursements_amt"
```

Creating an average debt per student and school rank plot

I do not really see any trends in this

```r
attach(Ranking_Tuition)
plot(Rank, (Total_Loan_amt/Enrollment), xlab = "School Rank", ylab = "Average Student Loan")
```




```r
detach(Ranking_Tuition)
```

Summary Statistics

```r
Ranking_Tuition <- Ranking_Tuition %>%
  mutate(Avg_Loan = Total_Loan_amt / Enrollment)

summary(Ranking_Tuition$Rank)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0    37.5    83.0    77.6   118.2   148.0
```

```r
summary(Ranking_Tuition$Tuition)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   17112   36805   49784   46488   57278   63000
```

```r
summary(Ranking_Tuition$Avg_Loan)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    5466  127328  191063  266941  374302  928663
```

```r
sd(Ranking_Tuition$Rank)
```

```
## [1] 45.613
```

```r
sd(Ranking_Tuition$Tuition)
```

```
## [1] 12410.2
```

```r
sd(Ranking_Tuition$Avg_Loan)
```

```
## [1] 192843
```

Mean tuition is about 46,488 with moderate spread (sd = 12,410). Loans range from 5,466 to 928,663. Mean average loan is 266,941 but median is $191,063 (right-skewed distribution).

Correlation Analysis with Rank and Tuition

```r
cor.test(Ranking_Tuition$Rank, Ranking_Tuition$Avg_Loan)
```

```
##
##  Pearson's product-moment correlation
##
## data:  Ranking_Tuition$Rank and Ranking_Tuition$Avg_Loan
## t = -0.88864, df = 98, p-value = 0.3764
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.2808916  0.1089247
## sample estimates:
##         cor
## -0.08940621
```

```r
cor.test(Ranking_Tuition$Tuition, Ranking_Tuition$Avg_Loan)
```

```
##
##  Pearson's product-moment correlation
##
## data:  Ranking_Tuition$Tuition and Ranking_Tuition$Avg_Loan
## t = 3.945, df = 98, p-value = 0.0001501
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1873985 0.5282019
## sample estimates:
##       cor
## 0.3701905
```

Rank vs. Average Loan: correlation is -.089 (very weak negative correlation). p-value is 0.3764 which is not significant. There is no strong linear relationship between school rank and average student loan.

Tuition vs. Average Loan: correlation is 0.370 (moderate positive). p-value is 0.00015 which is very significant. Higher tuition tends to be associated with higher student loans.

Linear Regression Model

```r
# Model 1: Avg Loan ~ Rank
model_rank <- lm(Avg_Loan ~ Rank, data = Ranking_Tuition)
summary(model_rank)
```

```
##
## Call:
## lm(formula = Avg_Loan ~ Rank, data = Ranking_Tuition)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -290430 -136644  -71876   95662  683797
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 296273.0    38238.9   7.748 8.72e-12 ***
## Rank          -378.0      425.4  -0.889    0.376
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 193000 on 98 degrees of freedom
## Multiple R-squared:  0.007993,   Adjusted R-squared:  -0.002129
## F-statistic: 0.7897 on 1 and 98 DF,  p-value: 0.3764
```

```r
# Model 2: Avg Loan ~ Tuition
model_tuition <- lm(Avg_Loan ~ Tuition, data = Ranking_Tuition)
summary(model_tuition)
```

```
##
## Call:
## lm(formula = Avg_Loan ~ Tuition, data = Ranking_Tuition)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -316249 -117479  -33298   64244  627991
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -478.534  70138.008  -0.007  0.99457
## Tuition         5.752      1.458   3.945  0.00015 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 180100 on 98 degrees of freedom
## Multiple R-squared:  0.137,  Adjusted R-squared:  0.1282
## F-statistic: 15.56 on 1 and 98 DF,  p-value: 0.0001501
```

```r
# Model 3: Avg Loan ~ Rank + Tuition
model_both <- lm(Avg_Loan ~ Rank + Tuition, data = Ranking_Tuition)
summary(model_both)
```

```
##
## Call:
```

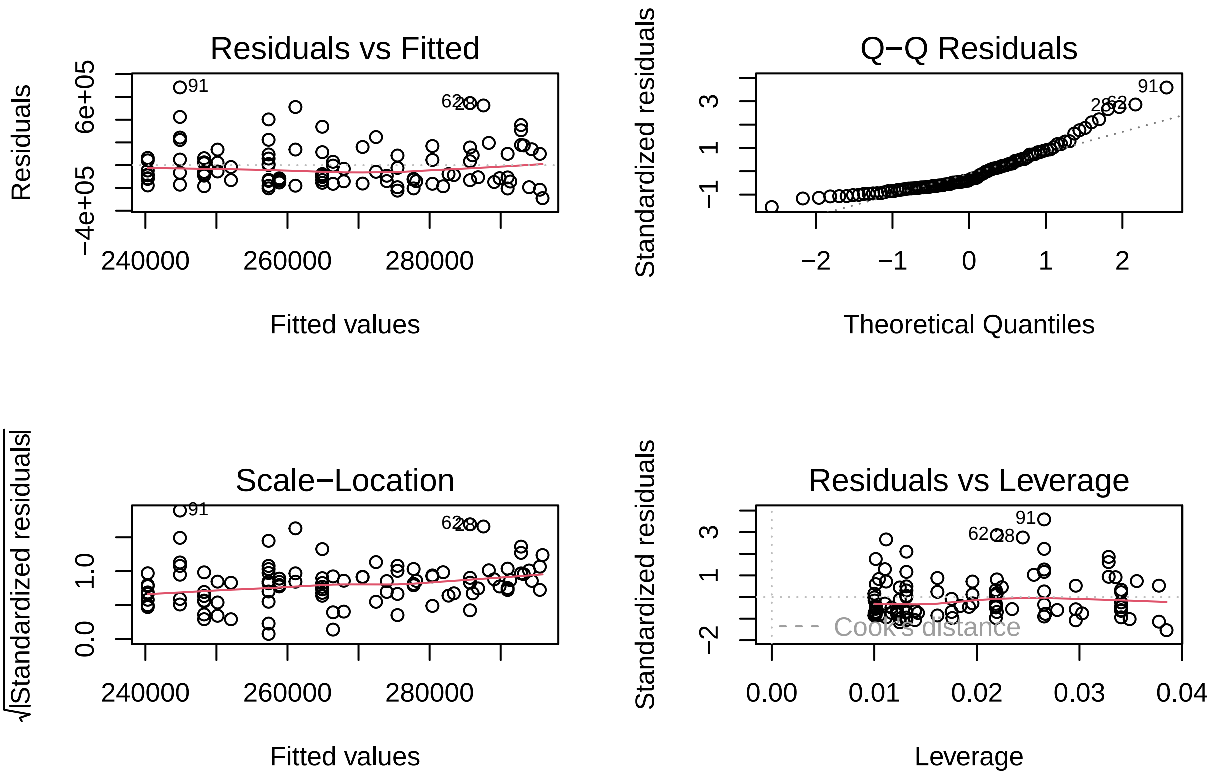
```
## lm(formula = Avg_Loan ~ Rank + Tuition, data = Ranking_Tuition)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -266235 -117881  -30251   68598  563446
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.641e+05  1.131e+05  -1.452   0.1499
## Rank         9.031e+02  4.930e+02   1.832   0.0701 .
## Tuition      7.765e+00  1.812e+00   4.285 4.31e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 177900 on 97 degrees of freedom
## Multiple R-squared:  0.1659, Adjusted R-squared:  0.1487
## F-statistic: 9.646 on 2 and 97 DF,  p-value: 0.0001511
```

Avg Loan vs Rank: For each one-place increase in rank (e.g., going from #1 to #2), the average loan amount decreases by $378 on average. The p-value for rank is 0.376, so it is not significant. Since R^2 is 0.008, this means that 0.8% of the variation in average student loan is explained by Rank.

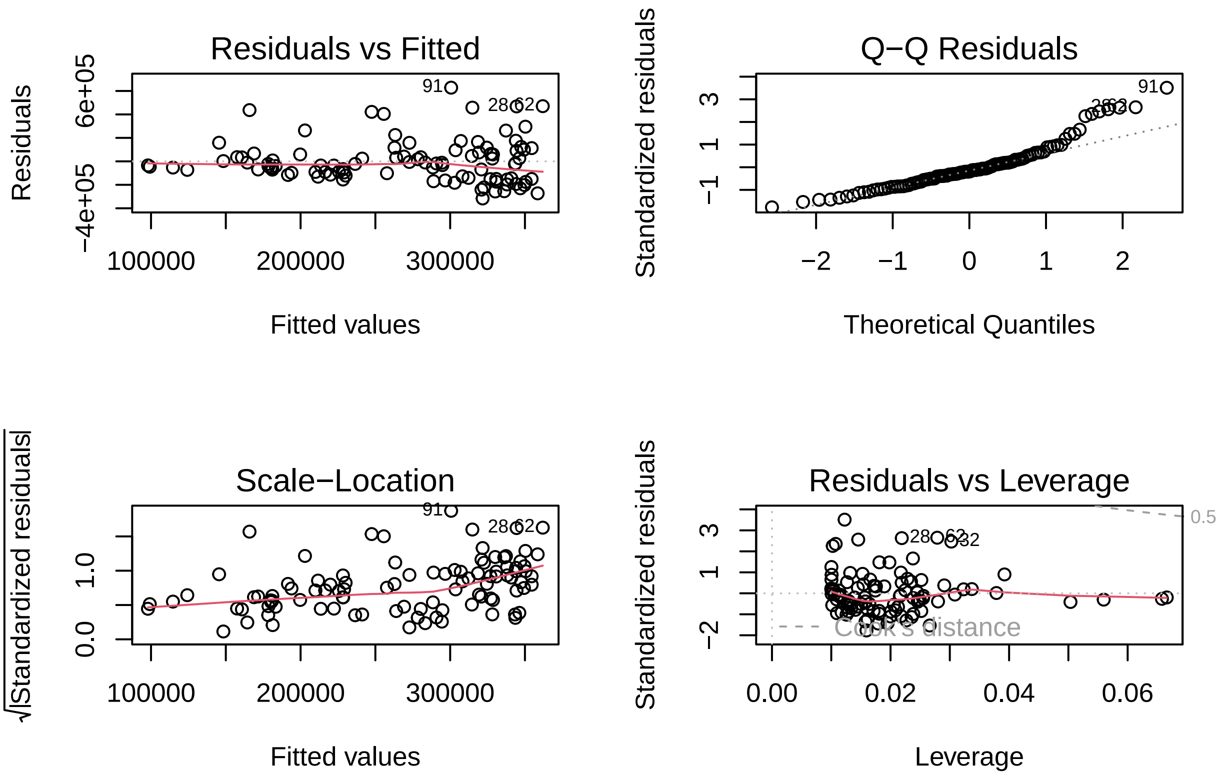
Avg Loan vs Tuition: For every 1 dollar increase in tuition, the average loan increases by about $5.75. R^2 is 0.137 whic means tuition explains about 13.7% of the variation in average loan. The p-value is very small which is strongly significant.

Avg Loan vs Rank and Tuition: Each 1-rank increase increases average loan by about $903 (p = 0.07 marginal significance). Each 1 dollar more in tuition increases average loan by 7.77 (strongly significant). Together, rank and tuition explain 16.6% of the variability. Tuition remains the stronger factor, but rank adds a small contribution to explaining loan amounts.

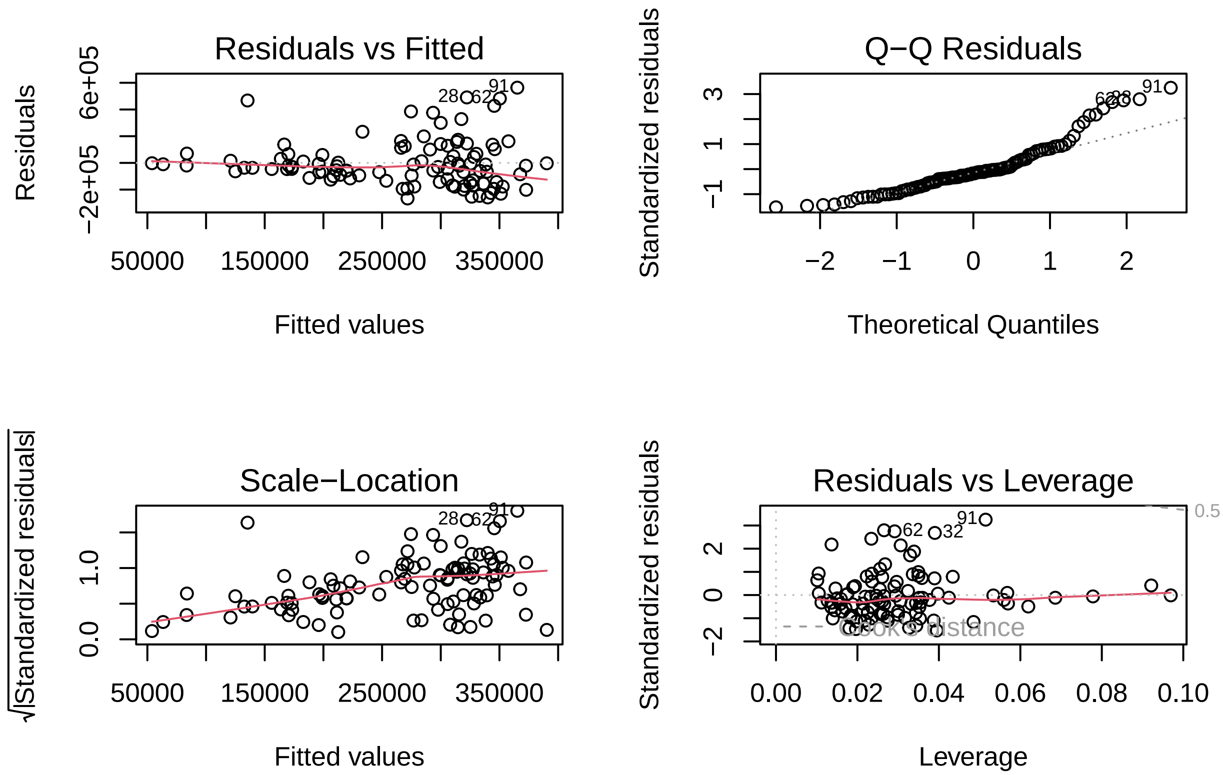
```r
par(mfrow = c(2,2))  # show 4 plots together
plot(model_rank)
```




```r
plot(model_tuition)
```




```
plot(model_both)
```




```r
par(mfrow = c(1,1))   # reset layout
```

Linearity: generally random, but some spread increases at higher fitted values. There is some mild curvature in Residuals vs Fitted. Normality of residuals: Small deviations and a little right-skewed Constant variance: Residual spread slightly increases at higher predicted loan amounts No extreme outliers. The rank + tuition model (3) provides the best fit.

Monte Carlo Simulation

```r
set.seed(123)
m <- 10000
sim_tuition <- runif(m, min = 20000, max = 70000)
sim_loans <- 0.1 * sim_tuition + rnorm(m, mean = 0, sd = 10000)

mean(sim_loans)
```

```
## [1] 4507.732
```

```r
sd(sim_loans) / sqrt(m)
```

```
## [1] 101.0368
```

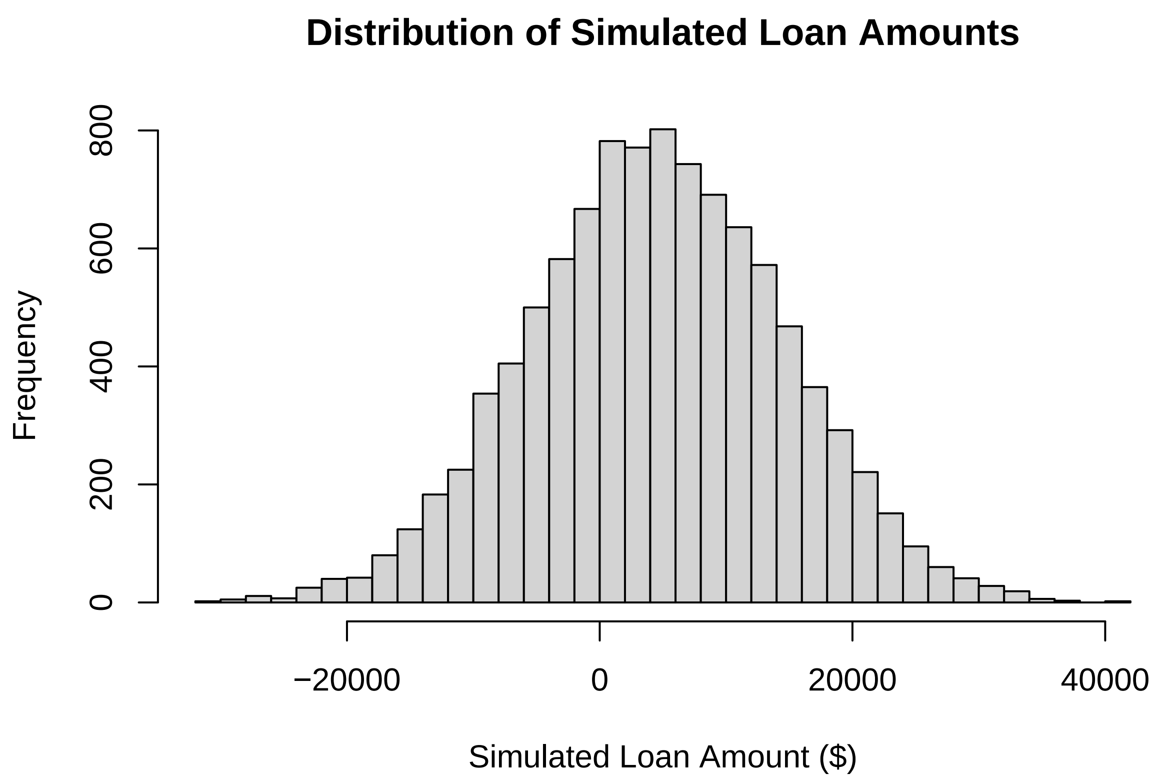
```r
mean(Ranking_Tuition$Avg_Loan)
```

```
## [1] 266940.8
```

The simulated average loan was about 4,508 but the real-world average loan from the dataset is 266,941. Real-world student loans are much higher than what random tuition simulations predict. This duggests that other factors also strongly contribute to student loan burdens.

Histogram of Simulated Loans Distribution from Monte Carlo

```
hist(sim_loans,
     breaks = 50,
     main = "Distribution of Simulated Loan Amounts",
     xlab = "Simulated Loan Amount ($)")
```



Prediction of average student loan for a school with ___ rank and ___ tuiton

```
# Can enter any value
rank_sim <- 30
tuition_sim <- 60000

new_school_sim <- data.frame(
  Rank = rank_sim,
  Tuition = tuition_sim
)

predict(model_both, newdata = new_school_sim, interval = "confidence")
```

```
##        fit    lwr      upr
## 1 328874.4 273801 383947.8
```

For this example, predicted avg loan is 328,874.40. This prediction can provide an estimate of the expected student loan burden for schools with similar characteristics. However, the model only uses rank and tuition. It doesn't account for financial aid, cost of living, family income, etc.

95% Confidence Intervals for Mean Loan

```
mean_loan <- mean(Ranking_Tuition$Avg_Loan)
se_loan <- sd(Ranking_Tuition$Avg_Loan) / sqrt(nrow(Ranking_Tuition))
lower_bound <- mean_loan - 1.96 * se_loan
upper_bound <- mean_loan + 1.96 * se_loan
c(lower_bound, upper_bound)
```

```
## [1] 229143.6 304738.0
```

We are 95% confident the true average loan is between [229143.6, 304738.0].

One sample t-test: is mean loan different from $25,000?

```
t.test(Ranking_Tuition$Avg_Loan, mu = 25000)
```

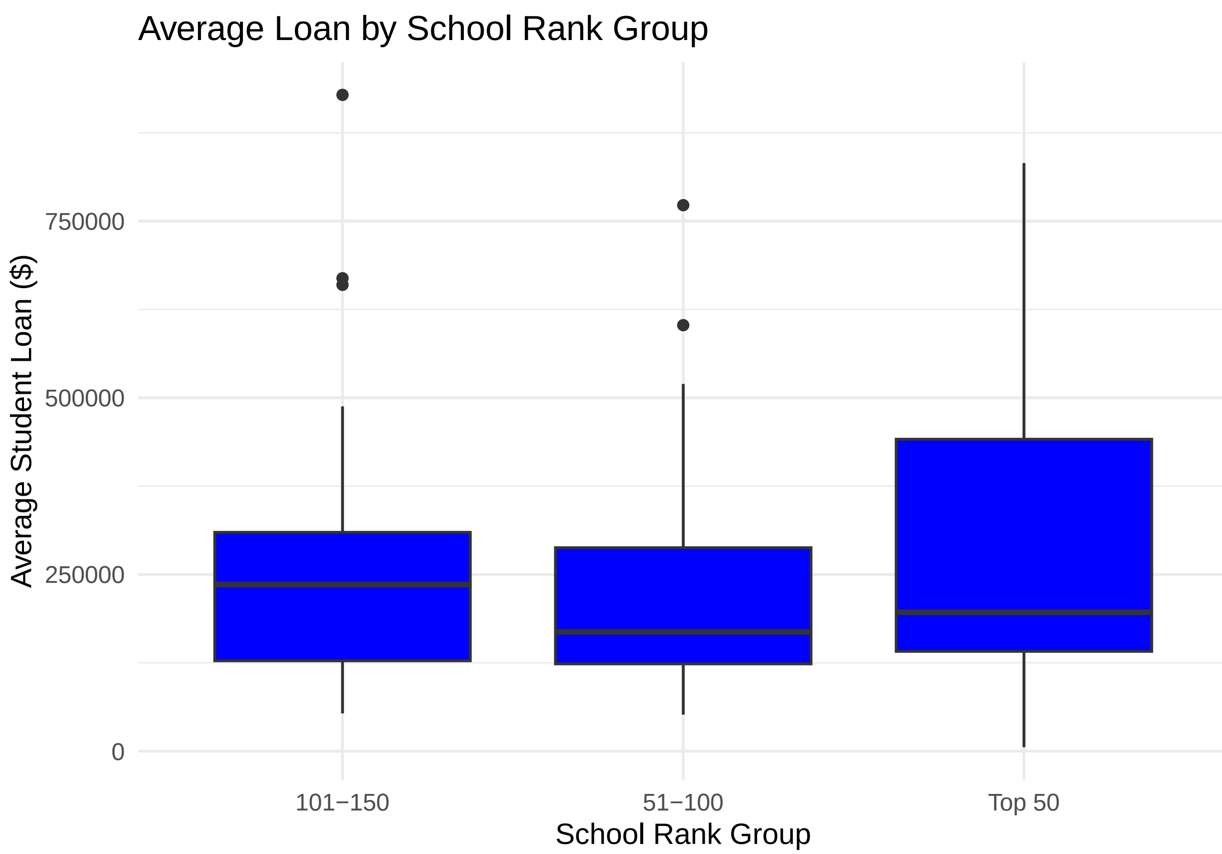
```
##
##  One Sample t-test
##
## data:  Ranking_Tuition$Avg_Loan
## t = 12.546, df = 99, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 25000
## 95 percent confidence interval:
##  228676.6 305205.0
## sample estimates:
## mean of x
##  266940.8
```

p value is very small, so we reject the null hypothesis and the mean loan is not 25,000.

Boxplot: Group schools by Rank Group

```
library(ggplot2)
Ranking_Tuition <- Ranking_Tuition %>%
  mutate(RankGroup = case_when(
    Rank <= 50 ~ "Top 50",
    Rank <= 100 ~ "51-100",
    Rank <= 150 ~ "101-150",
    TRUE ~ "151+"
  ))

ggplot(Ranking_Tuition, aes(x = RankGroup, y = Avg_Loan)) +
  geom_boxplot(fill = "blue") +
  labs(x = "School Rank Group", y = "Average Student Loan ($)",
       title = "Average Loan by School Rank Group") +
  theme_minimal()
```



Top 50 ranked schools show a wide range of loan amounts, with some students having extremely high debt. 51–100 ranked schools have lower average student loans and fewer outliers. 101–150 ranked schools show higher median loan amounts, but the spread is more contained. ‘